Nelson-Siegel

Jan Alsterlind

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1 Bond Pricing Using a Nelson–Siegel Spot Curve

1.1 General pricing framework

The price P of a fixed-coupon bond with cash flows $\{C_i\}$ at times $\{t_i\}$ is given by discounting each cash flow using the corresponding spot rate:

$$P = \sum_{i=1}^{N} C_i \, D(t_i), \tag{1}$$

where $D(t)$ is the discount function. Using continuously compounded spot rates $r(t)$:

$$D(t) = e^{-r(t)\,t}. \tag{2}$$

Thus,

$$P = \sum_{i=1}^{N} C_i \, e^{-r(t_i)t_i}. \tag{3}$$

1.2 Nelson–Siegel spot rate specification

The Nelson–Siegel (1987) parametrization models the instantaneous spot rate as:

$$r(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\tau}}{t/\tau} + \beta_2 \left(\frac{1 - e^{-t/\tau}}{t/\tau} - e^{-t/\tau} \right). \tag{4}$$

Interpretation:

- β_0: long-term level
- β_1: slope (short-term component)
- β_2: curvature (medium-term hump)
- τ: decay parameter controlling where the hump occurs

The discount function becomes:

$$D(t) = \exp\big[-t \cdot r(t) \big]. \tag{5}$$

1.3 Numerical example

Consider a 2-year annual coupon bond with:

- Face value: 100
- Coupon: 5% (so $C_1 = 5$, $C_2 = 105$)

Assume Nelson–Siegel parameters:

$$\beta_0 = 0.03, \quad \beta_1 = -0.02, \quad \beta_2 = 0.02, \quad \tau = 1.55.$$

Step 1: Compute spot rates

At $t = 1$:

$$r(1) = 0.03 - 0.02 \cdot \frac{1 - e^{-1/1.55}}{1/1.55} + 0.02 \cdot \left(\frac{1 - e^{-1/1.55}}{1/1.55} - e^{-1/1.55} \right). \tag{6}$$

Numerically:

$$r(1) \approx 0.0195.$$

At $t = 2$:

$$r(2) \approx 0.0245.$$

Step 2: Discount factors

$$D(1) = e^{-0.0195 \cdot 1} \approx 0.9807, \tag{7}$$

$$D(2) = e^{-0.0245 \cdot 2} \approx 0.9522. \tag{8}$$

Step 3: Price calculation

$$P = 5 \cdot 0.9807 + 105 \cdot 0.9522 \tag{9}$$

$$= 4.9035 + 99.9810 \tag{10}$$

$$= 104.88. \tag{11}$$

2 From model prices to yields and parameter estimation

The essential idea can be extended in two important ways. First, for any given model price P, one can compute the corresponding **yield to maturity** (YTM), denoted y, defined implicitly as the constant rate that satisfies:

$$P = \sum_{i=1}^{N} \frac{C_i}{(1+y)^{t_i}}. \tag{12}$$

This transformation allows model-implied prices to be expressed in terms of **model-implied yields**, which are often easier to compare with market quotations.

Second, if we observe a cross-section of bond market data (prices or yields), the Nelson–Siegel (or Svensson) parameters can be estimated by fitting the model to the data.

Let θ denote the vector of parameters (e.g. $\theta = (\beta_0, \beta_1, \beta_2, \tau)$). Two common approaches are:

Price-based estimation:

$$\min_{\theta} \sum_{j=1}^{M} \left(P_j^{\text{model}}(\theta) - P_j^{\text{market}} \right)^2, \tag{13}$$

Yield-based estimation:

$$\min_{\theta} \sum_{j=1}^{M} \left(y_j^{\text{model}}(\theta) - y_j^{\text{market}} \right)^2. \tag{14}$$

Here:

- $P_j^{\text{model}}(\theta)$ is obtained by discounting cash flows using the parametric spot curve
- $y_j^{\text{model}}(\theta)$ is the yield implied by that model price
- P_j^{market} and y_j^{market} are observed bond prices and yields

3 Why yield fitting and price fitting become (almost) equivalent with appropriate weighting

A useful way to understand the connection between price-based and yield-based estimation is to start from the classical first-order relationship between bond prices and yields. For a small change in yield dy, the corresponding relative change in price satisfies:

$$\frac{dP}{P} \approx -\text{Dur} \cdot dy, \tag{15}$$

where Dur denotes the (modified) duration of the bond. Multiplying both sides by P gives the absolute price change:

$$dP \approx -P \cdot \text{Dur} \cdot dy. \tag{16}$$

The quantity

$$\text{DV01} = \frac{\partial P}{\partial y} \tag{17}$$

is defined as the dollar value of a one basis point change in yield. Comparing with the expression above yields the standard identity:

$$\text{DV01} = P \cdot \text{Dur}. \tag{18}$$

Thus, price and yield errors are linked by the linear approximation:

$$\varepsilon_P \approx -\text{DV01} \cdot \varepsilon_y, \qquad \varepsilon_y \approx -\frac{\varepsilon_P}{\text{DV01}}. \tag{19}$$

A yield-based least squares estimator minimizes

$$\sum_{j=1}^{M} \varepsilon_{y,j}^2. \tag{20}$$

Using the approximation above, this becomes

$$\sum_{j=1}^{M} \left(\frac{\varepsilon_{P,j}}{\text{DV01}_j} \right)^2. \tag{21}$$

This shows that yield fitting is, to first order, equivalent to price fitting with weights proportional to $1/\text{DV01}^2$. In other words, the yield-based objective function corresponds exactly to:

$$\min_\theta \sum_{j=1}^{M} \left(\frac{P_j^{\text{model}}(\theta) - P_j^{\text{market}}}{\text{DV01}_j} \right)^2. \tag{22}$$

The implication is that the two estimation approaches differ only in how they weight instruments. Long-maturity bonds have large DV01 and therefore large price sensitivity to yield changes. Yield-based estimation automatically downweights these bonds in price space, because a given yield error corresponds to a large price error. Price fitting with $1/\text{DV01}^2$ weights reproduces exactly this normalization.

Consequently, yield fitting and appropriately weighted price fitting produce nearly identical parameter estimates, especially when yield changes are small and the duration approximation is accurate.

4 Conclusions

In summary, the Nelson–Siegel framework provides a unified approach where observed bond prices or yields are used to estimate a smooth parametric spot rate curve, which in turn generates consistent model prices and model yields across maturities.